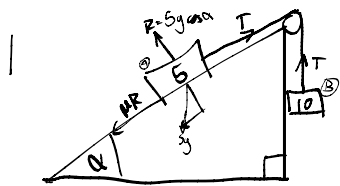


Fm 16 Forces 6 HW

Consolidation

$$a) \quad \textcircled{1} \quad T - g \cos \alpha - 5g \sin \alpha = 5a$$

$$10g - T = 10a$$

$$10g - g \cos \alpha - 5g \sin \alpha = 15a$$

$$10g - \frac{4}{5}g - 3g = 15a$$

$$a = \frac{31}{25}g$$

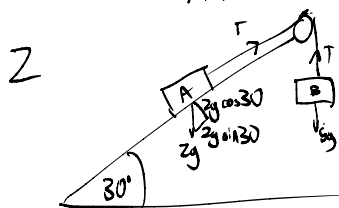
$$= 4.05$$

$$= 4.1 \text{ ms}^{-2}$$

$$b) \quad T = 10g - 10a$$

$$= \frac{88}{5}g$$

$$= 57N$$



$$c) \quad 5g - T = 5a$$

$$T - 2g \sin 30 = 2a$$

$$4g = 7a$$

$$a = \frac{4}{7}g$$

$$i) \quad T = \frac{8}{7}g + g$$

$$= \frac{15}{7}g = 21N$$

b) Tension is the same in all places

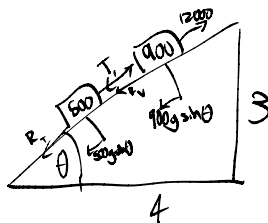
$$c) \quad F = \left| \begin{pmatrix} T \cos 30 \\ T + T \sin 30 \end{pmatrix} \right|$$

$$= \sqrt{\left(\frac{21}{2}\sqrt{3}\right)^2 + \left(21\frac{1}{2}\right)^2}$$

$$= 21\sqrt{3}$$

$$= \frac{15\sqrt{3}}{7}g$$

3



$$1200 - 900g \sin \theta - T - R_v = 400a$$

$$T - 500g \sin \theta - R_t = 600a$$

$$1200 - 1400g \sin \theta - R_t - R_v = 1400a$$

$$1200 - 840g - 2200 = 1400a$$

$$a = \frac{1065}{1400}$$

$$= 1.12 \text{ ms}^{-2}$$

$$b) \quad T = 500a + 500g \sin \theta + R_t$$

$$= 500 \times 1.12 + 500 \times 9.8 \times \frac{3}{5} + 600$$

$$= 4100$$

c) Maybe unrealistic
as acceleration $\uparrow$, $R \uparrow$

$$4) \quad T - 2g \sin 30 - \mu(2g \cos 30) = 5$$

$$3g - T = 7.5 \rightarrow T = 21.9N$$

$$b) \quad 3g - 2g \sin 30 - \mu(2g \cos 30) = 12.5$$

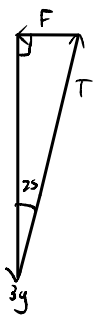
$$\mu = 0.42$$

$$c) \quad F = \left| \begin{pmatrix} T \cos 30 \\ T + T \sin 30 \end{pmatrix} \right|$$

$$= 38N$$

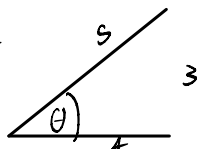
Exam Qs

1
a)



$$\begin{aligned} T \cos 25 &= 3g \\ T &= 32.4 \text{ N} \\ T \sin 25 &= F \\ F &= 13.7 \text{ N} \end{aligned}$$

2



b) $v^2 = u^2 + 2as$

$$1.2^2 = 2a$$

$$a = 0.48$$

c) $2.5g - T = 0.48 \times 2.5$

$$\begin{aligned} T &= 2.5g - 0.48 \times 2.5 \\ &= 23.3 \text{ N} \end{aligned}$$

d) $T - 2g \sin \theta - \mu 2g \cos \theta = 0.46$

$$\mu = 0.675$$

e) Time to hit ground: $1.5 = 0 + \frac{1}{2} \times 0.46 t^2$

$$t = 2.5$$

At $t = 2.5 \text{ s}$, $v_A = 0 + 2.5 \times 0.48 = 1.2 \text{ ms}^{-1}$

Then decel: $-2g \sin \theta - 2(0.675)g \cos \theta = 2a$
 $a = -11.172 \text{ ms}^{-2}$

$$v^2 = u^2 + 2as$$

$$0 = 1.2^2 + 2 \times -11.172 \times s$$

$$\begin{aligned} s &= \frac{-1.2^2}{2 \times -11.172} \\ &= 0.064 \text{ m} \end{aligned}$$